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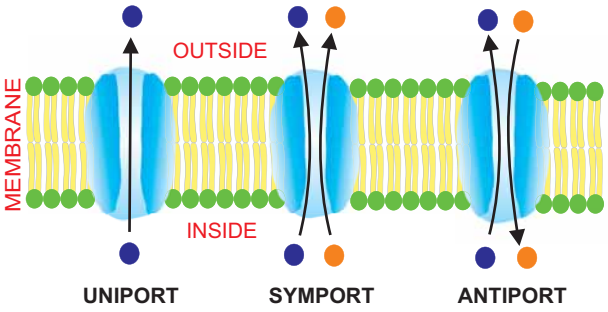


Figure 11.5: Direction of transport

transports two types of molecules across the membrane in the same direction.

iii. ***Antiport or Counter Transport***: An **antiport** is an integral membrane transport protein that simultaneously transports two different molecules, in opposite directions, across the membrane.

11.2.2 Active Transport

The main disadvantage of passive transport processes like diffusion is the lack of control over the transport of selective molecules. There is a possibility of harmful substances entering the cell by a concentration gradient in the diffusion process. But selective permeability of cell membrane has a great control over entry and exit of molecules. Active transport is the entry of molecules against a concentration gradient and an uphill process and it needs energy which comes from ATP. Passive transport uses kinetic energy of molecules moving down a gradient whereas, active transport uses cellular energy to move them against a gradient. The transport proteins discussed in facilitated diffusion can also transport ions or molecules against a concentration gradient with the expenditure of cellular energy as an active process. Pumps use a source of free energy such as ATP or light to drive the thermodynamically uphill transport of ions or molecules. The pump action is an example of active transport. Example: Na^+-K^+-ATPase pump (Table 11.1).

Table 11.1 Comparison of different transport mechanisms

Property	Passive transport		Active transport
	Simple diffusion	Facilitated diffusion	
Nature of process	Physical	Biological	Biological
Requirement for presence of membrane protein	No	Yes	Yes
Selectivity of molecule	No	Yes	Yes
Saturation of transport	No	Yes	Yes
Uphill transport	No	No	Yes
Energy requirement (ATP)	No	No	Yes
Sensitivity to inhibitors	No	Yes	Yes

Check your grasp!

What are the similarities and differences between co- transport and counter transport?

Solution:

Similarity: In both system two molecules are involved for the unidirectional transport.

Difference: In co-transport, two molecules are transported together whereas, in counter transport two molecules are transported in opposite direction to each other.